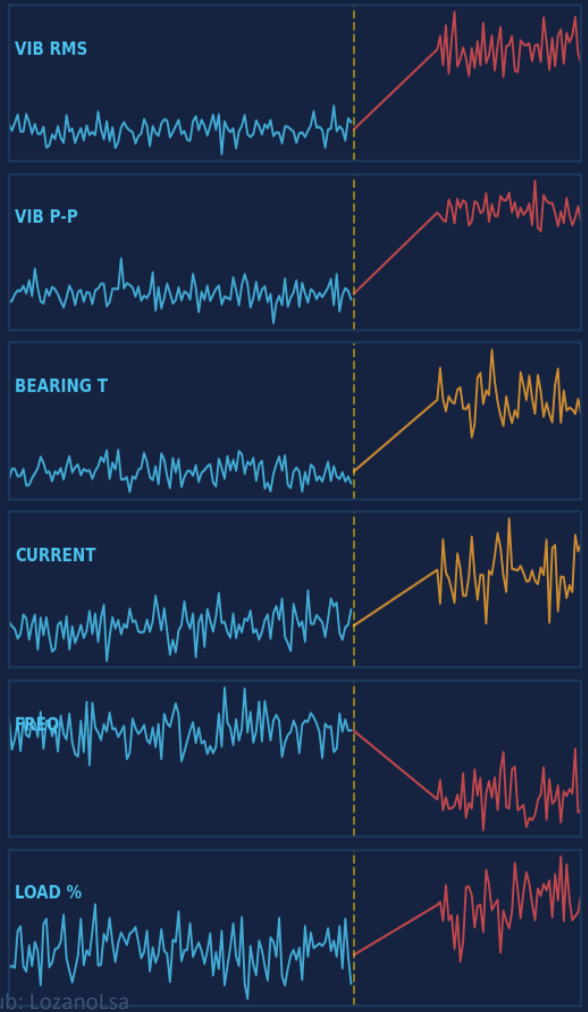
Motor Failures Are Not Random.

What if motor failure could be detected before the breakdown occurs?

K-Nearest Neighbors · Condition Monitoring · Predictive Maintenance



1,500
Motor Records

6
Sensor Signals

97.8%
Accuracy

0.996
ROC-AUC

The Real Problem

A motor does not fail suddenly — it degrades through measurable physical signals for weeks before shutdown.

Cost

Unplanned downtime, emergency repairs, lost production hours

Safety

Motor failures in critical systems create hazardous conditions

Operations

Cascading failures, maintenance bottlenecks, line disruptions

33.3% *of motor records show incipient failure signals — detectable in sensor data long before catastrophic breakdown.*

Every motor failure has a history. The signals are there. The question is whether we're reading them in time.

From Events to Probabilities

Old Thinking

Failures are sudden events:

"It tripped / it didn't."

Reactive maintenance.
Fix after breakdown.
No detection. No warning.
Just unplanned downtime.

New Thinking

Failure is a probabilistic state:

"How likely is incipient failure now?"

Score each motor continuously.
Flag degradation before shutdown.
Schedule maintenance proactively.
Convert signals into decisions.

The right question is not 'Did it fail?' — it's 'How likely is incipient failure, and which signals are driving it?'

The Data Behind the Problem

Six condition-monitoring signals — each one capturing a different physical dimension of motor degradation:

VIB RMS	**Vibration RMS (mm/s)** *Overall vibration energy — primary indicator of mechanical degradation. Normal: ~2.2 | Failure: ~4.3*
VIB P-P	**Vibration Peak-to-Peak (mm/s)** *Captures impact events: looseness, imbalance, bearing defects. Normal: ~5.0 | Failure: ~9.0*
BEARING T	**Bearing Temperature (°C)** *Friction buildup from lubrication loss or overloading. Normal: ~60°C | Failure: ~82°C*
CURRENT	**Motor Current (A)** *Overcurrent indicates increased mechanical resistance. Normal: ~28A | Failure: ~35A*
FREQ	**Dominant Frequency (Hz)** *Frequency drop signals misalignment or mechanical looseness. Normal: ~29.5Hz | Failure: ~24.5Hz*
LOAD %	**Load Percentage (%)** *Operating load context — high load amplifies all other degradation signals. Range: 41–111%*

KNN is ideal here — motor physics says that similar sensor profiles should share similar operating states.

Why K-Nearest Neighbors?

If two motors have similar vibration, temperature, and current profiles — they're likely in the same operating state.

Distance = Physics

KNN finds the k most similar historical motor states. Physical similarity maps directly to operational similarity — an intuition engineers share.

Non-linear Boundaries

Motor degradation creates irregular decision boundaries in sensor space. KNN adapts naturally — no linear assumption required.

Scaling is Essential

StandardScaler prevents Ampere-scale current from dominating mm/s vibration. Preprocessing inside Pipeline prevents data leakage.

GridSearchCV Tuning

Optimal k=5, weights='distance', p=1 (Manhattan). Distance weighting gives closer neighbors more diagnostic influence.

Distance-weighted KNN: the neighbors most similar to the current reading carry the most diagnostic weight.

Model Results

97.8%	0.996	96.6%	0.963
Accuracy	ROC-AUC	Recall (Failure)	F1 (Failure)

Permutation Importance (accuracy drop when shuffled)

vib_rms_mms	**Δ −8.2%**
vib_peak_to_peak_mms	**Δ −6.1%**
bearing_temp_c	**Δ −3.9%**
motor_current_a	Δ −1.4%
dominant_freq_hz	Δ −0.6%
load_pct	Δ −0.1%

Confusion Matrix (450 test records)

	Pred: Normal	Pred: Failure
Actual: Normal	**298 (TN)**	2 (FP)
Actual: Failure	5 (FN)	**145 (TP)**

Only 5 failures missed. In predictive maintenance, each FN represents an undetected degradation risk.

AUC 0.996 — the model nearly perfectly ranks failing motors above healthy ones across all thresholds.

From Insight to Action

VIB RMS Δ−8.2%	→ Primary alarm trigger. Set continuous monitoring threshold at 3.0 mm/s — anything above initiates inspection scheduling.
VIB P-P Δ−6.1%	→ Impact event detector. Rising peak-to-peak alongside stable RMS signals looseness or early bearing defects specifically.
Bearing T Δ−3.9%	→ Thermal confirmation signal. Above 72°C with elevated vibration = strong failure indicator. Use as secondary threshold.
Motor Current Δ−1.4%	→ Overcurrent amplifies mechanical signals. Combine with vibration to distinguish electrical from mechanical degradation.
Freq Δ−0.6%	→ Contextual signal — frequency drop confirms misalignment type. Low individual importance but useful for failure classification.

The model doesn't replace the engineer — it tells them which motor to inspect and why, before the shift ends.

What-If Simulator — Motor Condition Scorecard

Same model. Three motors. One shows the boundary — where the model switches from Normal to Failure.

Motor A — Healthy		Motor B — Failure		Motor C — Borderline	
Vib RMS	2.0 mm/s	Vib RMS	4.8 mm/s	Vib RMS	3.2 mm/s
Vib P-P	5.0 mm/s	Vib P-P	9.8 mm/s	Vib P-P	6.5 mm/s
Bearing	60°C	Bearing	88°C	Bearing	68°C
Current	28A	Current	37A	Current	30A
Freq	30 Hz	Freq	23 Hz	Freq	28.5 Hz
Load	70%	Load	92%	Load	78%
2% failure probability		**98% failure probability**		**45% failure probability**	

Motor C insight: elevated vibration at 3.2 mm/s is sufficient to place this motor at 45% — continuous monitoring is warranted.

Probability is a guide, not a verdict. The simulator turns model output into shift-level maintenance decisions.

This project is not about predicting motor failures.

It is about catching them while there's still time.

KNN doesn't predict the future —
it finds the motors whose present looks like past failures.

Feel free to download the full project: dataset, notebook, and simulator.

Change k. Try different distance metrics. Adjust the thresholds.

Because machine reliability has no finish line.